IMDB SQL PROJECT

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INTRODUCTION

The IMDB Movies Dataset is a real-world collection of film industry data that includes detailed information about movies, directors, budgets, revenues, ratings, and popularity scores. This dataset provides valuable insights into how movies perform commercially and critically across global audiences.

In this project, SQL (Structured Query Language) is used as the primary tool to extract, filter, and analyze data from the database. Through structured queries, the project evaluates movie performance based on profitability, audience popularity, and ratings while also examining the impact of directors on overall movie success. The analysis helps identify the most successful films, the most profitable directors, and emerging trends within the entertainment industry.

IMDB Movies Dataset is a real-world database containing details of movies, directors, and financial performance. This project uses SQL queries to analyze movie performance, profitability, popularity, and director impact on success.

Overall, this project demonstrates how SQL can be effectively used to transform raw data into meaningful business intelligence, supporting better decision-making in film production, marketing, and investment strategies.

OBJECTIVES

The primary goal of this project is to analyze the IMDB movie dataset using SQL in order to extract meaningful insights related to movie performance, profitability, director influence, and market trends. The project aims to utilize database querying techniques to support data-driven decision-making in the entertainment industry.

The specific objectives of this project are:

1. To analyze and identify the most popular movies based on audience engagement metrics such as popularity scores and vote counts.
2. To evaluate the profitability of movies by calculating financial indicators such as revenue, budget, and net profit in order to understand the commercial success of films.
3. To identify the most successful and influential directors by analyzing the number of movies directed and the total profit generated by their movies.
4. To study the gender distribution of directors in the dataset and assess the level of female representation in the film industry.
5. To examine the relationship between budget and revenue to determine whether higher investment leads to higher financial return.
6. To analyze movie ratings and vote averages to evaluate audience satisfaction and movie quality over time.
7. To filter and compare movies released after the year 2000 in order to understand recent industry trends and performance patterns.
8. To apply SQL queries such as JOIN, GROUP BY, ORDER BY, COUNT, and SUM for effective data extraction and analysis.
9. To convert raw data into structured information that can be used for business planning, marketing strategy, and investment decisions.
10. To develop strong practical knowledge of SQL through real-world database analysis and reporting.
11. To provide insights that can help production houses and streaming platforms select profitable content and reliable directors.

DATABASE ARCHITECTURE

The database used in this project is named **project_movie_database**, which is designed as a relational database to store and manage movie-related information in a structured format. The database consists of two main tables: **Movies** and **Directors**, which are logically connected to support detailed analysis.

1. Movies Table

The **Movies** table stores all information related to individual movies. Each record in this table represents one movie. It contains attributes such as the movie title, budget, revenue, popularity score, release date, ratings, and the unique identifier of the director associated with the movie.

This table plays a key role in financial and performance analysis, as it allows the system to:

- Track revenue and budget for profitability analysis
- Measure audience interest through popularity scores
- Analyze viewer ratings and votes
- Compare movies across different time periods

2. Directors Table

The **Directors** table contains information related to film directors. Each record in this table represents one director, including their name, gender, and department.

This table is used to:

- Identify directors involved in movie production
- Perform gender-based analysis
- Evaluate director productivity and performance
- Connect directors to multiple movies they have directed

DATA DESCRIPTION

Movies Table

The **Movies table** contains detailed information related to individual movies and their performance. Each record represents a unique movie along with its business and audience-related attributes. This table is the primary source for financial and popularity analysis in this project.

The key attributes in the Movies table include:

- **Title / Original Title** – The official name of the movie.
- **Budget** – The amount spent on producing the movie.
- **Revenue** – The total money earned by the movie globally.
- **Popularity** – A numerical score indicating viewer interest and demand.
- **Release Date** – The official date on which the movie was released.
- **Vote Average** – The average rating given by viewers.
- **Vote Count** – The total number of votes received by the movie.
- **Overview & Tagline** – Short descriptions and promotional summaries of each movie.
- **Director ID** – A reference linking the movie to its director.

Directors Table

The **Directors table** contains personal and professional information related to film directors involved in movie production.

The main attributes include:

- **ID** – A unique identifier for each director.
- **Name** – The full name of the director.
- **Gender** – Indicates the gender of the director (Male / Female).
- **Department** – Specifies the area of work such as directing or production.

KEY PERFORMANCE INDICATORS

Key Performance Indicators (KPIs) are measurable values used to evaluate the success and performance of movies and directors based on financial results, audience engagement, and ratings. In this project, KPIs are used to understand movie performance and support business and marketing decisions.

The KPIs used in this project are:

1. Total Movies

This KPI counts the total number of movies available in the database. It helps in understanding the size of the dataset and provides an overview of the analysis scope.

2. Profit

Profit is calculated using the formula:

Profit = Revenue – Budget

This KPI measures the financial success of a movie or director. High profit indicates strong market performance and return on investment, while low or negative profit indicates losses.

3. Popularity

Popularity reflects how much attention a movie has received from audiences. Higher popularity scores indicate more audience interest, higher search traffic, and stronger public engagement.

This KPI is useful for:

- Marketing analysis
- Trend detection
- Measuring fan interest

4. Vote Average

Vote Average shows the average rating given by users. It represents the quality perception of a movie based on audience feedback.

This KPI helps:

- Evaluate movie quality
- Compare audience satisfaction levels
- Identify critically successful films

5. Director Success

Director success is measured by:

- Number of movies directed
- Total profit generated
- Average ratings of movies

This KPI helps identify:

- Top-performing directors
- Industry leaders
- Reliable professionals for future projects

ANALYSIS

1.Join Both Tables.

```
SELECT *
FROM movies m
LEFT JOIN directors d
ON m.director_id = d.id
union
SELECT *
FROM movies m
RIGHT JOIN directors d
ON m.director_id = d.id;
```

2. Can you get all data about movies.

```
Select * From movies;
```

3. How do you get all data about directors.

```
select * from directors;
```

4. Check how many movies are present in IMDB.

```
select count(original_title) as total_movies from movies;
```

5. Find these 3 directors: James Cameron ; Luc Besson ; John Woo

```
SELECT *
FROM directors
WHERE name IN ('James Cameron', 'Luc Besson', 'John Woo');
```

6. Find all directors with name starting with S.

```
SELECT count(*) AS Start_with_S
FROM directors
WHERE name LIKE 'S%'
```

7. Count female directors.

```
select count(*) as females from directors where gender = 1;
```

8. Find the name of the 10th first women directors

```
select name  from directors where gender = 1 limit 10;
```

9. What are the 3 most popular movies.

```
select original_title from movies order by popularity desc limit 3;
```

10. What are the 3 most bankable movies

```
select original_title ,(revenue - budget) as profit from movies order by
profit desc limit 3;
```

11.What is the most awarded average vote since the January 1st, 2000

```
select original_title,
        vote_average from movies
        where release_date > "2000-1-1"
        order by vote_average desc;
```

12. Which movie(s) were directed by Brenda Chapman.

```
Select
         m.original_title
        from movies m join directors d
```

```
on m.director_id = d.id
where d.name ="Brenda Chapman";
```

```
SELECT m.title
FROM movies m
JOIN directors d
ON m.director_id = d.id
WHERE d.name = 'Brenda Chapman';
```

13. Which director made the most movies

```
SELECT d.name,
 COUNT(m.id) AS movie_count
FROM movies m
JOIN directors d
ON m.director_id = d.id
 GROUP BY d.name
 ORDER BY movie_count DESC
LIMIT 1;
```

14. Which director is the most bankable

```
select d.name,
sum(m.revenue - m.budget) as profit
 from movies m
 join directors d
on m.director_id = d.id
group by d.name order by profit desc limit 3;
```

CONCLUSION

This project demonstrates how SQL can effectively transform raw movie data into meaningful business insights through structured queries and analytical techniques. By working with the IMDB dataset, the project successfully analyzed movie performance based on profitability, popularity, audience ratings, and director contributions.

The analysis helped in identifying the most successful movies and directors, understanding revenue trends, and observing audience preferences. The study also highlighted the low representation of female directors in the industry, revealing an important area for improvement and inclusivity in the film sector.

Furthermore, the project proves that SQL is a powerful tool for handling large datasets, performing complex calculations, and generating accurate insights that support business decision-making. The ability to work with relationships between tables and apply aggregation functions such as SUM, COUNT, and GROUP BY enhances analytical capability and ensures reliable reporting.

Overall, this project provides valuable insights for filmmakers, investors, and marketing teams by supporting strategic planning, financial forecasting, and data-driven decision-making in the entertainment industry. It also strengthens practical SQL skills and demonstrates how database management plays a crucial role in modern data analytics.

THANK YOU